H6 Solving differential equations A level only

Name: ____________________

Class: ____________________

Date: ____________________

Time: **387 minutes**

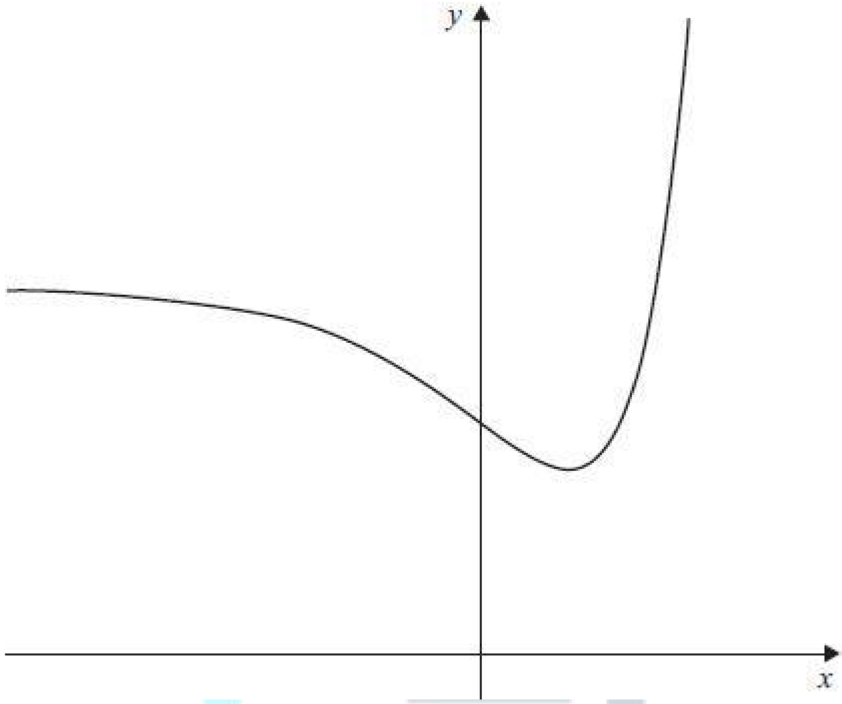
Marks: **322 marks**

Comments:

Q1.

A function f has domain $\mathbb{R}$ and range $\{y \in \mathbb{R} : y \geq \mathrm{e}\}$

The graph of $y = \mathrm{f}(x)$ is shown.



The gradient of the curve at the point (x, y) is given by $\frac{\mathrm{d}y}{\mathrm{d}x} = (x - 1)\mathrm{e}^x$

Find an expression for $\mathrm{f}(x)$.

Fully justify your answer.

(Total 8 marks)

Q2.

A market trader notices that daily sales are dependent on two variables:

number of hours, t, after the stall opens

total sales, x, in pounds since the stall opened.

The trader models the rate of sales as directly proportional to $\frac{8 - t}{x}$

After two hours the rate of sales is £72 per hour and total sales are £336

(a) Show that

$$x\frac{\mathrm{d}x}{\mathrm{d}t} = 4032(8-t)$$

(3)

(b) Hence, show that

$$x^2 = 4032t(16 - t)$$

(3)

(c) The stall opens at 09.30.

(i) The trader closes the stall when the rate of sales falls below £24 per hour.

Using the results in parts (a) and (b), calculate the earliest time that the trader closes the stall.

(6)

(ii) Explain why the model used by the trader is not valid at 09.30.

(2)

(Total 14 marks)

Q3.

The height, x metres, of a column of water in a fountain display satisfies the differential equation $\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{8\sin 2t}{3\sqrt{x}}$, where t is the time in seconds after the display begins.

(a) Solve the differential equation, given that initially the column of water has zero height.
Express your answer in the form $x = \mathrm{f}(t)$

(7)

(b) Find the maximum height of the column of water, giving your answer to the nearest cm.

(1)

(Total 8 marks)

Q4.

(a) Find $\int x\sqrt{x^2+3}\,\mathrm{d}x$.

(2)

(b) Solve the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x\sqrt{x^2+3}}{\mathrm{e}^{2y}}$$

given that $y = 0$ when $x = 1$. Give your answer in the form $y = \mathrm{f}(x)$.

(7)

(Total 9 marks)

Q5.

(a) Find $\int t\cos\left(\frac{\pi}{4}t\right) \mathrm{d}t$.

(4)

(b) The platform of a theme park ride oscillates vertically. For the first 75 seconds of the ride,

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{t\cos\left(\frac{\pi}{4}t\right)}{32x}$$

where x metres is the height of the platform above the ground after time t seconds.

At t = 0, the height of the platform above the ground is 4 metres.

Find the height of the platform after 45 seconds, giving your answer to the nearest centimetre.

(6)

(Total 10 marks)

Q6.

Solve the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} = y^2 x\sin 3x$$

given that y = 1 when $x = \frac{\pi}{6}$. Give your answer in the form $y = \frac{9}{\mathrm{f}(x)}$.

(Total 9 marks)

Q7.

(a) A water tank has a height of 2 metres. The depth of the water in the tank is h metres at time t minutes after water begins to enter the tank. The rate at which the depth of the water in the tank increases is proportional to the difference between the height of the tank and the depth of the water.

Write down a differential equation in the variables h and t and a positive constant k.

(You are not required to solve your differential equation.)

(3)

(b) (i) Another water tank is filling in such a way that t minutes after the water is turned on, the depth of the water, x metres, increases according to the differential equation

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{1}{15x\sqrt{2x-1}}$$

The depth of the water is 1 metre when the water is first turned on.

Solve this differential equation to find t as a function of x.

(8)

(ii) Calculate the time taken for the depth of the water in the tank to reach 2 metres, giving your answer to the nearest 0.1 of a minute.

(1)

(Total 12 marks)

Q8.

(a) (i) Solve the differential equation $\frac{dx}{dt} = \sqrt{x}\sin\left(\frac{t}{2}\right)$ to find x in terms of t.

(3)

(ii) Given that $x = 1$ when $t = 0$, show that the solution can be written as

$$x = (a - \cos bt)^2$$

where a and b are constants to be found.

(3)

(b) The height, x metres, above the ground of a car in a fairground ride at time t seconds is

modelled by the differential equation $\frac{dx}{dt} = \sqrt{x}\sin\left(\frac{t}{2}\right)$.

The car is 1 metre above the ground when $t = 0$.

(i) Find the greatest height above the ground reached by the car during the ride.

(2)

(ii) Find the value of t when the car is first 5 metres above the ground, giving your answer to one decimal place.

(2)

(Total 10 marks)

Q9.

A giant snowball is melting. The snowball can be modelled as a sphere whose surface area is decreasing at a constant rate with respect to time. The surface area of the sphere is A cm^2 at time t days after it begins to melt.

(a) Write down a differential equation in terms of the variables A and t and a constant k, where $k > 0$, to model the melting snowball.

(2)

(b) (i) Initially, the radius of the snowball is 60 cm, and 9 days later, the radius has halved.

Show that $A = 1200\pi(12 - t)$.

(You may assume that the surface area of a sphere is given by $A = 4\pi r^2$, where r is the radius.)

(4)

(ii) Use this model to find the number of days that it takes the snowball to melt completely.

(1)

(Total 7 marks)

Q10.

(a) Express $\dfrac{1}{(3-2x)(1-x)^2}$ in the form $\dfrac{A}{3-2x} + \dfrac{B}{1-x} + \dfrac{C}{(1-x)^2}$.

(4)

(b) Solve the differential equation

$$\frac{dy}{dx} = \frac{2\sqrt{y}}{(3-2x)(1-x)^2}$$

where $y = 0$ when $x = 0$, expressing your answer in the form

$$y^p = q\ln[f(x)] + \frac{x}{1-x}$$

where p and q are constants.

(9)

(Total 13 marks)

Q11.

Solve the differential equation $\dfrac{dy}{dx} = \dfrac{1}{y}\cos\left(\dfrac{x}{3}\right)$, given that $y = 1$ when $x = \dfrac{\pi}{2}$.

Write your answer in the form $y^2 = f(x)$.

(Total 6 marks)

Q12.

(a) Solve the differential equation

$$\frac{dx}{dt} = -\frac{1}{5}(x+1)^{\frac{1}{2}}$$

given that $x = 80$ when $t = 0$. Give your answer in the form $x = f(t)$.

(6)

(b) A fungus is spreading on the surface of a wall. The proportion of the wall that is unaffected after time t hours is x%. The rate of change of x is modelled by the differential equation

$$\frac{dx}{dt} = -\frac{1}{5}(x+1)^{\frac{1}{2}}$$

At $t = 0$, the proportion of the wall that is unaffected is 80%. Find the proportion of the wall that will still be unaffected after 60 hours.

(2)

(c) A biologist proposes an alternative model for the rate at which the fungus is spreading on the wall. The total surface area of the wall is 9 m^2. The surface area that is **affected** at time t hours is A m^2. The biologist proposes that the rate of change of A is proportional to the product of the surface area that is affected and the surface area that is unaffected.

(i) Write down a differential equation for this model.

(You are not required to solve your differential equation.)

(2)

(ii) A solution of the differential equation for this model is given by

$$A = \frac{9}{1 + 4e^{-0.09t}}$$

Find the time taken for 50% of the area of the wall to be affected. Give your answer in hours to three significant figures.

(4)

(Total 14 marks)

Q13.

(a) A differential equation is given by $\frac{dx}{dt} = -kt\,e^{\frac{1}{2}x}$, where k is a positive constant.

(i) Solve the differential equation.

(3)

(ii) Hence, given that $x = 6$ when $t = 0$, show that $x = -2\ln\left(\frac{kt^2}{4} + e^{-3}\right)$.

(3)

(b) The population of a colony of insects is decreasing according to the model

$\frac{dx}{dt} = -kte^{\frac{1}{2}x}$, where x thousands is the number of insects in the colony after time t minutes. Initially, there were 6000 insects in the colony.

Given that k = 0.004, find:

(i) the population of the colony after 10 minutes, giving your answer to the nearest hundred;

(2)

(ii) the time after which there will be no insects left in the colony, giving your answer to the nearest 0.1 of a minute.

(2)

(Total 10 marks)

Q14.

(a) Solve the differential equation

$$\frac{dx}{dt} = \frac{150\cos 2t}{x}$$

given that x = 20 when $t = \frac{\pi}{4}$, giving your solution in the form $x^2 = f(t)$.

(6)

(b) The oscillations of a 'baby bouncy cradle' are modelled by the differential equation

$$\frac{dx}{dt} = \frac{150\cos 2t}{x}$$

where x cm is the height of the cradle above its base t seconds after the cradle begins to oscillate.

Given that the cradle is 20 cm above its base at time $t = \frac{\pi}{4}$ seconds, find:

(i) the height of the cradle above its base 13 seconds after it starts oscillating, giving your answer to the nearest millimetre;

(2)

(ii) the time at which the cradle will first be 11 cm above its base, giving your answer to the nearest tenth of a second.

(2)

(Total 10 marks)

Q15.

Solve the differential equation

$$\frac{dy}{dx} = \frac{3\cos 3x}{y}$$

given that $y = 2$ when $x = \frac{\pi}{2}$. Give your answer in the form $y^2 = f(x)$.

(Total 5 marks)

Q16.

(a) Express $\frac{2}{x^2 - 1}$ in the form $\frac{A}{x-1} + \frac{B}{x+1}$.

(3)

(b) Hence find $\int \frac{2}{x^2 - 1} dx$.

(2)

(c) Solve the differential equation $\frac{dy}{dx} = \frac{2y}{3(x^2 - 1)}$, given that $y = 1$ when $x = 3$.

Show that the solution can be written as $y^3 = \frac{2(x-1)}{x+1}$.

(5)

(Total 10 marks)

Q17.

(a) (i) Solve the differential equation $\frac{dy}{dt} = y \sin t$ to obtain y in terms of t.

(4)

(ii) Given that $y = 50$ when $t = \pi$, show that $y = 50e^{-(1 + \cos t)}$.

(3)

(b) A wave machine at a leisure pool produces waves. The height of the water, y cm, above a fixed point at time t seconds is given by the differential equation

$$\frac{dy}{dt} = y \sin t$$

(i) Given that this height is 50 cm after π seconds, find, to the nearest centimetre, the heigh tof the water after 6 seconds.

(2)

(ii) Find $\frac{d^2 y}{dt^2}$ and hence verify that the water reaches a maximum height after π

seconds.

(4)
(Total 13 marks)

Q18.

(a) Solve the differential equation

$$\frac{dy}{dx} = \frac{\sqrt{1+2y}}{x^2}$$

given that $y = 4$ when $x = 1$.

(6)

(b) Show that the solution can be written as $y = \frac{1}{2}\left(15 - \frac{8}{x} + \frac{1}{x^2}\right)$

(2)
(Total 8 marks)

Q19.

(a) Solve the differential equation

$$\frac{dx}{dt} = -2(x-6)^{\frac{1}{2}}$$

to find t in terms of x, given that $x = 70$ when $t = 0$.

(6)

(b) Liquid fuel is stored in a tank. At time t minutes, the depth of fuel in the tank is x cm. Initially there is a depth of 70 cm of fuel in the tank. There is a tap 6 cm above the bottom of the tank. The flow of fuel out of the tank is modelled by the differential equation

$$\frac{dx}{dt} = -2(x-6)^{\frac{1}{2}}$$

(i) Explain what happens when $x = 6$.

(1)

(ii) Find how long it will take for the depth of fuel to fall from 70 cm to 22 cm.

(2)
(Total 9 marks)

Q20.

Solve the differential equation

$$\frac{dy}{dx} = 6xy^2$$

given that y = 1 when x = 2. Give your answer in the form y = f (x).

(Total 6 marks)

Q21.

(a) A cup of coffee is cooling down in a room. At time t minutes after the coffee is made, its temperature is x °C, where

$$x = 15 + 70e^{-\frac{t}{40}}$$

(i) Find the temperature of the coffee when it is made.

(1)

(ii) Find the temperature of the coffee 30 minutes after it is made.

(2)

(iii) Find how long it will take for the coffee to cool down to 60 °C.

(3)

(b) (i) Use integration to solve the differential equation

$$\frac{dx}{dt} = -\frac{1}{40}(x - 15), \qquad x > 15$$

given that x = 85 when t = 0, expressing t in terms of x.

(6)

(ii) Hence show $x = 15 + 70e^{-\frac{t}{40}}$

(2)

(Total 14 marks)

Q22.

A particle, of mass 12 kg, is moving along a straight horizontal line. At time t seconds, the particle has speed v m s^{-1}. As the particle moves, it experiences a resistance force of magnitude $4v^{\frac{1}{3}}$. No other horizontal force acts on the particle.

The initial speed of the particle is 8 m s^{-1}.

(a) Show that

$$v = \left(4 - \frac{2}{9}t\right)^{\frac{3}{2}}$$

(6)

(b) Find the value of t when the particle comes to rest.

(1)
(Total 7 marks)

Q23.

A car accelerates from rest along a straight horizontal road.

The car's engine produces a constant horizontal force of magnitude 4000 N.

At time t seconds, the speed of the car is v m s^{-1}, and a resistance force of magnitude $40v$ newtons acts upon the car.

The mass of the car is 1600 kg.

(a) Show that $\frac{dv}{dt} = \frac{100 - v}{40}$.

(2)

(b) Find the velocity of the car at time t.

(6)
(Total 8 marks)

Q24.

Alice places a toy, of mass 0.4 kg, on a slope. The toy is set in motion with an initial velocity of 1 m s^{-1} down the slope. The resultant force acting on the toy is $(2 - 4v)$ newtons, where v m s^{-1} is the toy's velocity at time t seconds after it is set in motion.

(a) Show that $\frac{dv}{dt} = -10\,(v - 0.5)$.

(2)

(b) By using $\int \frac{1}{v-0.5}\,dv = -\int 10\,dt$, find v in terms of t.

(5)

(c) Find the time taken for the toy's velocity to reduce to 0.55 m s^{-1}.

(3)
(Total 10 marks)

Q25.

A stone, of mass 5 kg, is projected vertically downwards, in a viscous liquid, with an initial speed of 7 m s^{-1}.

At time t seconds after it is projected, the stone has speed v m s^{-1} and it experiences a resistance force of magnitude $9.8v$ newtons.

(a) When $t \geq 0$, show that

$$\frac{dv}{dt} = -1.96(v - 5)$$

(2)

(b) Find v in terms of t.

(5)
(Total 7 marks)

Q26.

Vicky has mass 65 kg and is skydiving. She steps out of a helicopter and falls vertically. She then waits a short period of time before opening her parachute. The parachute opens at time $t = 0$ when her speed is 19.6 m s^{-1}, and she then experiences an air resistance force of magnitude $260v$ newtons, where v m s^{-1} is her speed at time t seconds.

(a) When $t > 0$:

(i) show that the resultant downward force acting on Vicky is

$$65(9.8 - 4v) \text{ newtons}$$

(1)

(ii) show that $\frac{dv}{dt} = -4(v - 2.45)$.

(2)

(b) By showing that $\int \frac{1}{v - 2.45} dv = -\int 4dt$, find v in terms of t.

(5)
(Total 8 marks)

Q27.

A car, of mass m kg , is moving along a straight horizontal road. At time t seconds, the car has speed v m s^{-1} . As the car moves, it experiences a resistance force of magnitude $2mv^{\frac{5}{4}}$ newtons. No other horizontal force acts on the car.

(a) Show that

$$\frac{dv}{dt} = -2v^{\frac{5}{4}}$$

(1)

(b) The initial speed of the car is 16 m s^{-1}.

Show that

$$v = \left(\frac{2}{t+1}\right)^4$$

(5)
(Total 6 marks)

Q28.

A golf ball, of mass m kg, is moving in a straight line across smooth horizontal ground. At time t seconds, the golf ball has speed v m s^{-1}. As the golf ball moves, it experiences a resistance force of magnitude $0.2mv^{\frac{1}{2}}$ newtons until it comes to rest. No other horizontal force acts on the golf ball.

Model the golf ball as a particle.

(a) Show that

$$\frac{dv}{dt} = -0.2v^{\frac{1}{2}}$$

(1)

(b) When $t = 0$, the speed of the golf ball is 16 m s^{-1}.

Show that $v = (4 - 0.1t)^2$.

(5)

(c) Find the value of t when $v = 1$.

(3)

(d) Find the distance travelled by the golf ball as its speed decreases from 16 m s^{-1} to 1 m s^{-1}.

(4)
(Total 13 marks)

Q29.

A particle is moving along a straight line. At time t, the velocity of the particle is v.

The acceleration of the particle throughout the motion is $-\frac{\lambda}{v^{\frac{1}{4}}}$, where λ is a positive constant. The velocity of the particle is u when $t = 0$.

Find v in terms of u, λ and t.

(Total 7 marks)

Q30.

A stone, of mass 0.05 kg, is moving along the smooth horizontal floor of a tank, which is filled with oil. At time t, the stone has speed v. As the stone moves, it experiences a resistance force of magnitude 0.08 v^2.

(a) Show that

$$\frac{dv}{dt} = -1.6v^2$$

(2)

(b) The initial speed of the stone is 3 m s^{-1}.

Show that

$$v = \frac{15}{5 + 24t}$$

(5)
(Total 7 marks)

Q31.

A stone, of mass m, is moving in a straight line along smooth horizontal ground.

At time t, the stone has speed v. As the stone moves, it experiences a total resistance force of magnitude $\lambda m v^{\frac{3}{2}}$, where λ is a constant. No other horizontal force acts on the stone.

(a) Show that

$$\frac{dv}{dt} = -\lambda v^{\frac{3}{2}}$$

(2)

(b) The initial speed of the stone is 9 m s^{-1}.

Show that

$$v = \frac{36}{(2 + 3\lambda t)^2}$$

(7)

(c) Find, in terms of λ, the time taken for the speed of the stone to drop to 4 m s^{-1}.

(3)
(Total 12 marks)

Q32.

A car, of mass m, is moving along a straight smooth horizontal road. At time t, the car has speed v. As the car moves, it experiences a resistance force of magnitude $0.05mv$. No other horizontal force acts on the car.

(a) Show that

$$\frac{dv}{dt} = -0.05v$$

(1)

(b) When $t = 0$, the speed of the car is 20 m s^{-1}.

Show that $v = 20e^{-0.05t}$.

(4)

(c) Find the time taken for the speed of the car to reduce to 10 m s^{-1}.

(3)

(d) Find, in terms of m, the work done by the force in slowing the car from 20 m s^{-1} to 10 m s^{-1}.

(3)
(Total 11 marks)

Q33.

A car, of mass m, is moving along a straight smooth horizontal road. At time t, the car has speed v. As the car moves, it experiences a resistance force of magnitude $0.05mv$. No other horizontal force acts on the car.

(a) Show that

$$\frac{dv}{dt} = -0.05v$$

(1)

(b) When $t = 0$, the speed of the car is 20 m s^{-1}.

Show that $v = 20e^{-0.05t}$.

(4)

(c) Find the time taken for the speed of the car to reduce to 10 m s^{-1}.

(3)
(Total 8 marks)

Q34.

A stone of mass m is moving along the smooth horizontal floor of a tank which is filled with a viscous liquid. At time t, the stone has speed v. As the stone moves, it experiences a resistance force of magnitude λmv, where λ is a constant.

(a) Show that

$$\frac{dv}{dt} = -\lambda v$$

(2)

(b) The initial speed of the stone is U.

Show that

$$v = Ue^{-\lambda t}$$

(4)
(Total 6 marks)

Q35.

A car, of mass 1600 kg, is travelling along a straight horizontal road at a speed of 20 m s^{-1} when the driving force is removed. The car then freewheels and experiences a resistance force. The resistance force has magnitude $40v$ newtons, where v m s^{-1} is the speed of the car after it has been freewheeling for t seconds.

Find an expression for v in terms of t.

(Total 7 marks)